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阅读 C

海豹能感知血液中的氧气

说明文 · 动物科学 · 全文 400 词 · 逐句精读

¹ All **mammals** **take in** **oxygen** from the air they **breathe in**, and **release** CO₂ from their **blood** when **breathing out**.

所有哺乳动物都从吸入的空气中获取氧气，在呼气时把血液中的二氧化碳排出。

mammal n. 哺乳动物

take in 摄入；吸收

oxygen n. 氧气

breathe in 吸入

release v. 释放

blood n. 血液

breathe out 呼气

² Most **mammals** (哺乳动物) can't **directly** **detect** oxygen levels in the blood **supplied** to their **tissues**.

大多数哺乳动物无法直接察觉输送到身体组织的血液中的氧含量。

mammal n. 哺乳动物

directly adv. 直接地

detect v. 察觉；探测

supply v. 供应

tissue n. (生物) 组织

³ **Instead**, they **rely on** the rising level of CO₂ in their blood to **signal** that they might need to **take a breath**.

相反，它们依靠血液中不断升高的二氧化碳水平发出的信号，得知自己可能需要呼吸了。

instead adv. 相反

rely on 依靠

signal v. 发出信号

take a breath 吸一口气

⁴ But a **recent** study **published** in Science **suggests** **seals** (海豹) can **sense** the amount of oxygen in the blood, and change their diving **behavior** **in response**.

但发表在《科学》杂志上的一项最新研究表明，海豹能感知血液中的氧气含量，并据此改变潜水行为。

recent adj. 最近的

publish v. 发表

suggest v. 表明

seal n. 海豹

sense v. 感知

behavior n. 行为

in response 作为反应；相应地

⁵ To **find out** if oxygen levels **affected** seal behavior, **Professor** McKnight at the University of St. Andrews and his **colleagues** created a special **section** in a **pool** where young seals were held.

为了弄清氧气水平是否会影响海豹的行为，圣安德鲁斯大学的麦克奈特教授和同事在饲养幼年海豹的水池中专门隔出了一块区域。

find out 弄清

affect v. 影响

professor n. 教授

colleague n. 同事

section n. 区域；部分

pool n. 水池

⁶ In one **corner**, there was a breathing **chamber** (呼吸室), where they were **sheltered** from the rain and the wind.

在一个角落里有一个呼吸室，海豹在那里可以躲避风雨。

corner n. 角落

chamber n. 室

shelter v. 庇护；遮蔽

7 The breathing chamber **was surrounded by** **panels** that **prevented** **surface** swimming, yet swimming below the surface for about 200 feet would give the seals **access** to a **feeder** where they could eat as much fish as they liked.

呼吸室四周围着挡板，使海豹无法在水面游动；但如果在水下潜游大约200英尺，海豹就能到达一个投食点，在那里想吃多少鱼就吃多少。

be surrounded by 被……环绕

panel n. 挡板；面板

prevent v. 阻止

surface n. 水面

access n. （到达某处的）途径；接近的机会

feeder n. 投食器；投食点

⁸ Once the seals **got familiar with** the **setup**, the researchers started to **gradually** change the **composition** of the air in the breathing chamber, **increasing** or **reducing** the levels of oxygen and CO₂ to see an **effect** on their behavior.

海豹熟悉了这一实验布置后，研究人员开始逐渐改变呼吸室内的空气成分，提高或降低氧气和二氧化碳的含量，看看这对它们的行为有什么影响。

get familiar with 熟悉

setup n. 布置；装置

gradually adv. 逐渐地

composition n. 成分

increase v. 增加

reduce v. 减少

effect n. 影响

⁹ Sure enough, the higher the level of oxygen, the longer the seals stayed at the feeder.

果然，氧气含量越高，海豹在投食点停留的时间就越长。

sure enough 果然

the higher...the longer 越……越

¹⁰ The **finding** suggests that seals don't just **physically** **respond to** oxygen levels by changing their **heart rate** or breathing, but are also **sufficiently** **aware** of them to change their behavior.

这一发现表明，海豹不只是在生理上通过改变心率或呼吸来对氧气水平做出反应，而且能足够清楚地察觉到氧气水平，从而改变自己的行为。

finding n. 发现

physically adv. 生理上；身体上

respond to 对……做出反应

heart rate 心率

sufficiently adv. 充分地

aware adj. 意识到的

¹¹ This **ability** would put seals **in a class beyond** any land mammals that have been **tested**.

这种能力会让海豹超越所有接受过测试的陆地哺乳动物。

ability n. 能力

in a class beyond 比……高出一个层次；胜过……

test v. 测试

¹² Since oxygen levels on land **remain stable**, humans don't seem to have **evolved** (演化) to **notice** low blood oxygen levels, sometimes not even when they're **about to pass out** in **free-diving**.

由于陆地上的氧气水平保持稳定，人类似乎并没有进化出察觉血氧过低的能力，有时甚至在自由潜水中即将昏厥时也察觉不到。

remain v. 保持

stable adj. 稳定的

evolve v. 进化

notice v. 注意到

be about to 即将

pass out 昏厥

free-diving n. 自由潜水

¹³ Therefore, in free-diving without oxygen tanks, accidents are quite common.

因此，在不带氧气瓶的自由潜水中，事故相当常见。

therefore adv. 因此

tank n. （储气的）罐；瓶

accident n. 事故

¹⁴ Our **reliance** on sensing CO₂ levels in our blood **instead of** oxygen may **be to blame**.

我们依靠感知血液中的二氧化碳水平而不是氧气水平，这或许就是原因所在。

reliance n. 依赖

instead of 而不是

be to blame 应负责任；是原因

15 Actually, this is a perfectly reasonable strategy on land, where growing CO₂ tends to signal breathing issues.

其实，在陆地上这是一种完全合理的策略，因为在陆地上二氧化碳升高往往意味着呼吸出了问题。

actually adv. 实际上

perfectly adv. 完全地

reasonable adj. 合理的

strategy n. 策略

issue n. 问题

¹⁶ But when we **hold our breath** during diving, relying on CO₂ levels is **risky**, **especially** on **repeated** dives.

但当我们潜水屏住呼吸时，依靠二氧化碳水平是有风险的，尤其是在反复潜水时。

hold one's breath 屏住呼吸

risky adj. 有风险的

especially adv. 尤其

repeated adj. 反复的

17 This is because every time we **surface** and breathe in, our **sensitivity** to CO₂ is **decreased**, **even if** its levels are already high, and this increases the **chance** that a person will, without **awareness**, pass out before they get to the surface.

这是因为每次我们浮出水面吸气时，我们对二氧化碳的敏感度就会降低，即使它的含量已经很高；这就增加了一个人在毫无察觉的情况下，在到达水面之前昏厥的可能性。

surface v. 浮出水面

sensitivity n. 敏感度

decrease v. 降低

even if 即使

chance n. 可能性

awareness n. 察觉；意识